SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT3 — CASE STUDY

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO2 Statement:	*Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)*		
Student Name:	Tejaswini P	**Reg. No.:**	192324046
Section / Batch:	2023/ B.tech AIDS	**Date:**	28/7/2026

Code of Conduct

I Tejaswini *P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Tejaswini

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	**25%**	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.
Application	**20%**	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with well-reasoned conclusions.	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.
Organization	**15%**	The response is clearly structured, addressing every sub-part (a–d) in a logical sequence with	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		coherent overall flow.			
Accuracy	**10%**	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Problem 4 (10 Marks)

Scenario: A dataset tracks a company employees by City (nominal), Tenure in Years (numeric), and Performance Rating (ordinal: Low, Medium, High).

a) Design a single visualization that encodes all three variables simultaneously. Specify each aesthetic mapping.

Visualization: Bar Chart

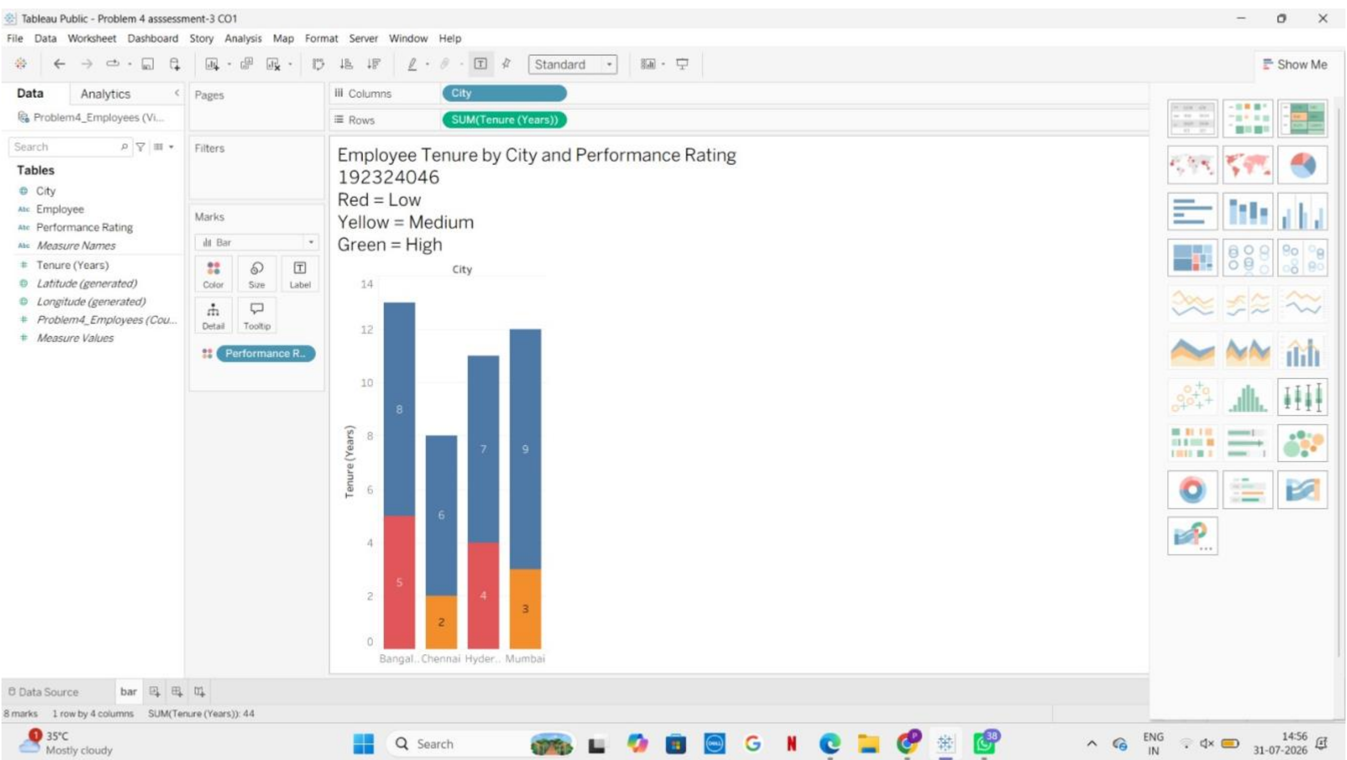
b) Which chart type did you choose, and why is it suited to combining a nominal, a numeric, and an ordinal variable?

I chose a Bar Chart because it effectively compares numeric values (Tenure) across categories (City). Using color to represent the ordinal Performance Rating allows all three variables to be displayed in a single, easy-to-understand visualization.

c) What is one limitation of your design that could cause confusion for a viewer, and how would you mitigate it?

Limitation: If there are many employees in the same city, the chart may become cluttered and difficult to read.

Mitigation: Use grouped bars or interactive filtering, and include a clear legend for the performance rating colors to improve readability.



Problem 14 (10 Marks)

Scenario: A quality control team measures a manufacturing defect rate (in parts per million, ranging from 1 to 100,000) across 12 production lines.

a) Design a chart to compare defect rates across the 12 lines, specifying your choice of axis scale.

Visualization: Bar Chart

b) Justify why a logarithmic axis may be necessary here, referencing the wide range of values.

A logarithmic axis is necessary because the defect rates span a very large range (1–100,000 PPM). On a normal (linear) scale, small values would appear almost invisible compared to very large values. A log scale compresses the larger values and expands the smaller ones, making all production lines easier to compare.

c) What potential misinterpretation risk exists if the audience is unfamiliar with reading logarithmic axes, and how would you mitigate it (e.g., through annotation)?

Risk: Viewers may incorrectly assume that equal distances on the axis represent equal numerical differences, leading to misinterpretation of the defect rates.

Mitigation: Clearly label the Y-axis as "Logarithmic Scale (PPM)", use tick marks such as 1, 10, 100, 1,000, 10,000, 100,000, and include a short note explaining that each step represents a 10× increase, helping the audience interpret the chart correctly.